

TRank-IoT

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Chapter 1

Introduction

The Internet of Things (IoT)[15] has witnessed exponential growth over the past decade, evolving from basic sensor networks to sophisticated systems that integrate smart cities, industrial automation, healthcare, and environmental monitoring[3]. These advancements have led to a massive deployment of IoT devices[19] in Low-power and Lossy Networks (LLNs), where resource constraints like limited battery power, memory, and bandwidth are prominent. Efficient communication among these devices is essential, and for this purpose, IPv6[12] Routing Protocol (RPL) was introduced by the **Internet Engineering Task Force** (IETF) [11] to address the specific routing needs of such networks.

The IPv6 Routing Protocol for Low Power and Lossy Networks (RPL)[2] is a distance vector routing protocol specifically designed for low-power devices operating under the **IEEE 802.15.4 standard** [11], incorporating the 6LoWPAN adaptation layer. Introduced by the Routing Over Low-Power and Lossy Networks working group, RPL addresses the unique routing requirements of Low Power and Lossy Networks (LLNs) by accounting for the constraints of energy, processing, and memory resources. This protocol facilitates communication among numerous nodes arranged in either a peer-to-peer or extended star topology[18].

RPL establishes a **multi-hop hierarchical structure**, where each node transmits data to its parent, which then forwards it until it reaches a sink or gateway node. The sink can send unicast messages to specific nodes within the network. Designed for efficiency in managing data routing among resource-constrained devices.

RPL [2]ensures bidirectional connectivity, robustness, and scalability through its hierarchical framework, efficient control message management, and flexible objective functions. Given the resource limitations of sensor nodes in Internet of Things (IoT)[15] applications, RPL has emerged as the standard routing protocol for **IPv6-based 6LoWPAN networks**, supporting a diverse array of IoT applications while addressing the need for optimal energy usage and computational efficiency.

RPL control Message Flow

RPL is used with the 6LoWPAN protocol to connect nodes via DODAG. It relies on four types of control messages[22]:

- DODAG Information Solicitation (DIS)
- DODAG Information Object (DIO)
- Destination Advertisement Object (DAO)
- Destination Advertisement Object Acknowledgment (DAO ACK)

The Flow of control messages are shown in figure 1.1 The DIO message helps other nodes join the DODAG network. The root node initiates the transmission of the DIO, and nodes that become part of the DODAG propagates DIO messages to other nodes. The DIO message is periodically transmitted and can also be triggered by DIS messages. It contains important information required for the construction and maintenance of the topology, such as the RPL Instance ID, DODAG ID, version number, and rank. RPL uses a trickle timer to manage the frequency of DIO messages[11], ensuring efficient and energy-saving communication.

When a device joins the DODAG and selects a preferred parent, it sends a DAO control message to the parent node. This message is then forwarded towards the root node. DAO messages[19] move upwards toward the root node, opposite to the downward movement of DIO messages. The root node processes the information contained in the DAO message to recognize the node joining the DODAG.

In storing mode, the parent node appends its DAO information to the packet received from the child node before forwarding it to the root node. In non-storing mode, the DAO packet is immediately forwarded to the root node upon receipt from the child node. DAO ACK messages, which are optional, are sent from the root node back to leaf nodes in a downward direction. Initially, the root node establishes the DODAG topology by calculating the required information and multicasts the DIO message to neighboring nodes. Neighboring nodes select their parent based on the Objective Function (OF) [16] and compute their rank value[5].

Each preferred parent acts as a router between the child node and the root node. The root node serves as the preferred parent for all initial nodes in the network. Nodes that are one hop away from the root node transmit DIO messages downwards, while DAO messages[6] are sent in response to the received DIO messages.

Type of RPL Base Attack

The Routing Protocol for Low-Power and Lossy Networks (RPL) is vulnerable to both external and internal attacks, as depicted in Figure 1.2. External attacks such as RFID

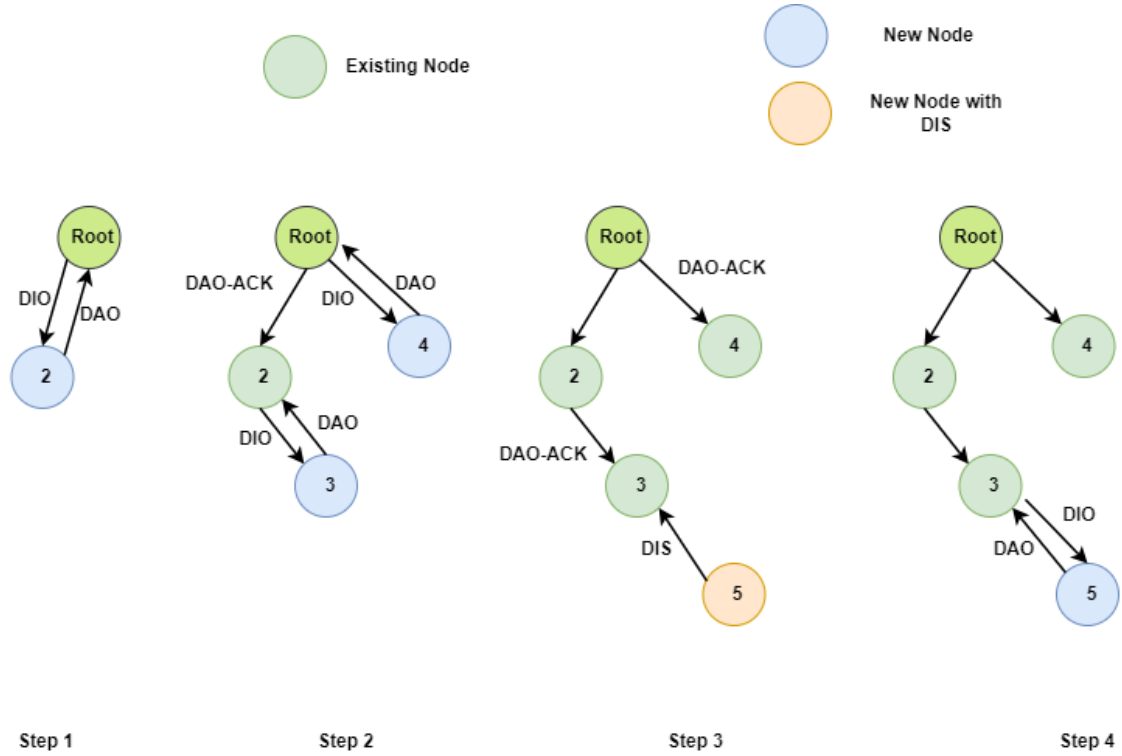


Figure 1.1: Control Messages Flow

tracking, eavesdropping, Distributed Denial of Service (DDoS), and Man-in-the-Middle (MITM) attacks are introduced by malicious nodes from outside the network. These attacks can intercept, reroute, or manipulate data traffic, significantly disrupting communication and causing data breaches.

On the other hand, internal attacks are more covert and dangerous since compromised nodes within the network manipulate the RPL protocol itself. One notable internal attack is the **rank attack**, where a malicious node advertises a falsified lower rank, tricking other nodes into selecting it as a parent.

This allows the attacker to control the traffic flow, degrade performance, and potentially cause network failure. Other significant internal threats include Sybil attacks, blackhole attacks, and sinkhole attacks, all of which disrupt the normal functioning of the network by exploiting the routing protocol to drop packets or manipulate routing decisions.

Rank Attack

In RPL (Routing Protocol for Low-Power and Lossy Networks)[15], rank is a metric used to define the position of a node relative to the root node (the destination). The rank of a node determines its distance from the root based on a variety of factors such as hop count or link quality. Nodes with lower ranks are closer to the root[6], and nodes with higher ranks are farther away.

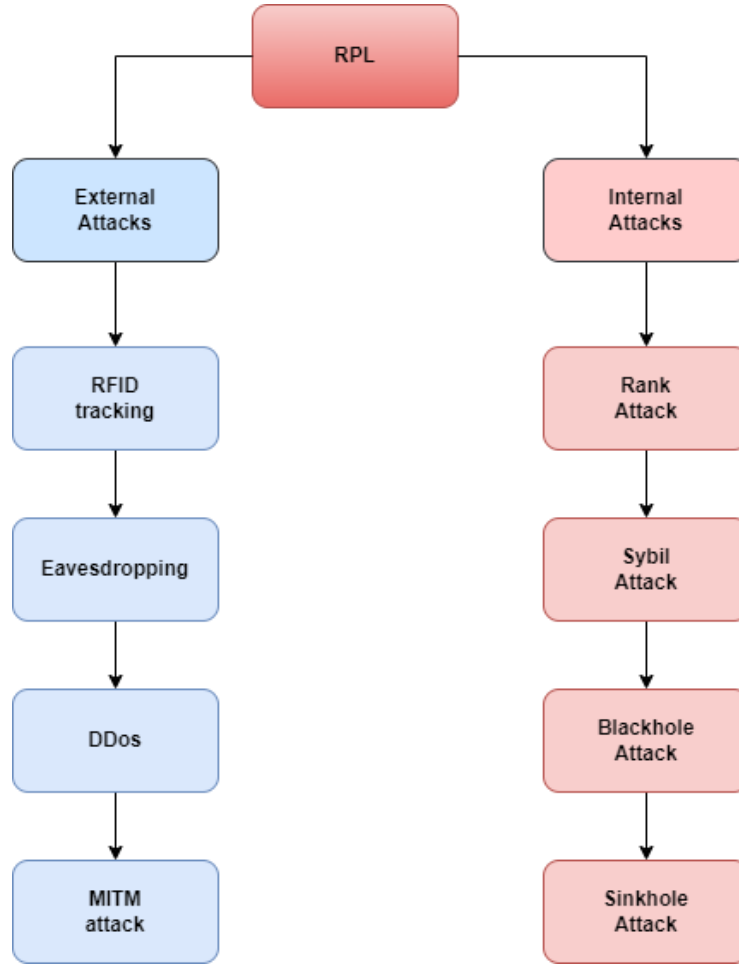


Figure 1.2: Type of RPL Attacks

This ranking system helps in building the network's tree structure, known as the Destination Oriented Directed Acyclic Graph (DODAG), where each node selects a parent with a lower rank to forward packets towards the root.

A Rank Attack [16] in RPL occurs when a malicious node manipulates its rank to disrupt the network's routing topology. Specifically, the malicious node advertises a lower rank than it should have, making it appear closer to the root node as shown in Figure 1.3.

In RPL, nodes select parents with lower ranks to establish the most efficient paths to the root. By falsely claiming a lower rank[5], the malicious node deceives neighboring nodes into choosing it as their preferred parent for forwarding data.

Once the malicious node is integrated into the network's routing structure, it can cause severe disruptions. These include creating routing loops where packets are forwarded in circles, leading to increased latency and energy consumption. Additionally, the attacker[20] can control a significant portion of the network's traffic and perform other malicious activities, such as dropping packets, delaying transmissions, or selectively forwarding data, further degrading network performance.

Rank attacks are particularly harmful because they distort the overall network topology[5] and often serve as a gateway for more complex attacks, like sinkhole and selective forwarding. By exploiting the RPL protocol's inability to validate the rank information of nodes, an attacker can manipulate routing decisions and severely impair the network's functionality.

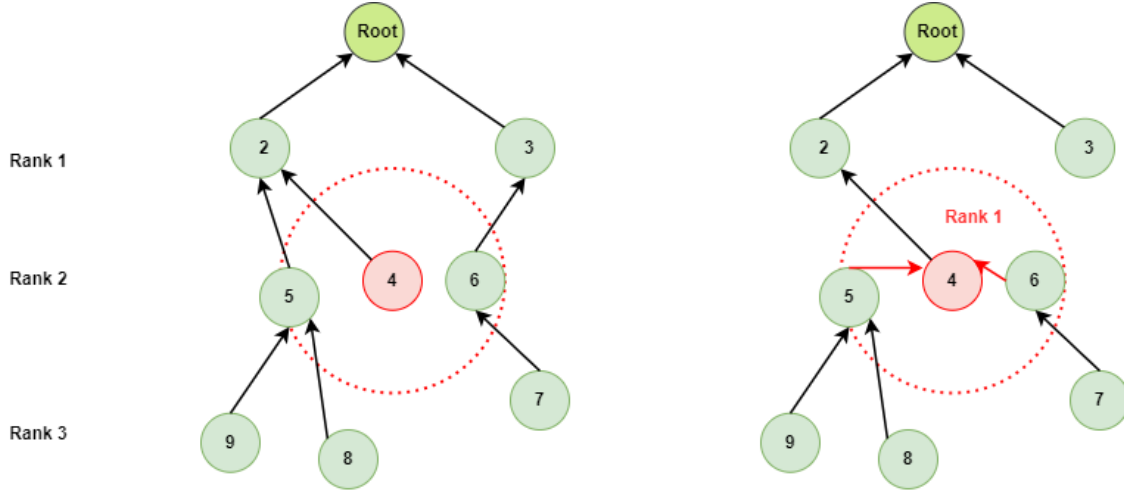


Figure 1.3: Rank Attack

My research focuses on addressing these internal vulnerabilities[20], particularly the rank attack, which remains one of the most disruptive forms of routing manipulation in RPL-based IoT[15] networks. While several solutions have been proposed, many are either too resource-intensive or fail to provide a lightweight, efficient defense suited to the limited resources of IoT devices.

1.1 Problem Domain

In a rank attack, malicious nodes manipulate their rank to deceive the routing protocol, leading to network disruptions as shown in 1.3 such as routing loops, increased packet loss, and energy depletion[17].

These attacks can degrade the overall performance of the IoT network, resulting in unreliable data transmission, reduced network lifespan, and potential security breaches. Despite the significance of this threat, existing detection and mitigation mechanisms for rank attacks[19] are either too computationally expensive or ineffective in IoT's[15] resource-constrained environments.

Traditional security measures often rely on heavyweight cryptographic techniques or complex detection algorithms that are unsuitable for the low-power devices typical of IoT networks. Moreover, existing research on rank attack[20] detection lacks comprehensive

solutions that are both lightweight and efficient enough to be deployed in real-world IoT scenarios.

The absence of a robust, scalable, and lightweight mechanism to detect and mitigate rank attacks in RPL-based IoT[15] networks highlights a critical gap in the current IoT security landscape. There is a pressing need for a security solution that not only detects rank attacks with high accuracy but also operates within the constraints of IoT devices, ensuring minimal overhead and preserving the network's overall efficiency[17].

The TRank-IoT project aims to address this gap by proposing a trust-based, lightweight, and efficient rank attack detection mechanism for IoT [15]networks. The proposed solution leverages trust-based models[16], where the trustworthiness of each IoT node is assessed based on its behavior and interactions with other nodes in the network.

By continuously monitoring node behavior as shown in figure1.4 and assigning trust values, the system can detect malicious nodes engaging in rank attacks without relying on computationally intensive processes. Additionally, the TRank-IoT mechanism is designed to integrate seamlessly with existing IoT protocols, ensuring that it operates within the energy [17]and computational limitations of typical IoT[15] devices.

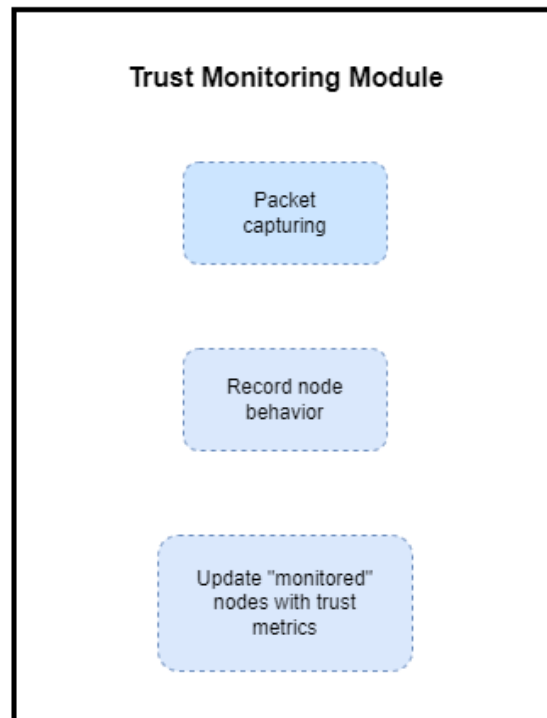


Figure 1.4: Trust Monitoring Module

1.2 Research Problem Statement

Existing security solutions for IoT[15] networks often rely on traditional methodologies that are either computationally intensive or fail to address the specific vulnerabilities inherent in RPL.

These approaches frequently overlook the potential of trust-based mechanisms[16] that evaluate the behavior and reliability of nodes within the network. The absence of effective detection and mitigation strategies tailored for rank attacks limits the overall resilience of IoT networks[20], making them susceptible to malicious exploitation.

This research seeks to develop a comprehensive framework for detecting and mitigating rank attacks[6] in RPL-based IoT networks through the implementation of a lightweight, trust-based mechanism. By focusing on the trustworthiness[16] of nodes and their interactions, the proposed solution aims to enhance detection accuracy while minimizing resource consumption.

The outcome of this research will provide valuable insights into improving the security and reliability of IoT networks[15], ultimately contributing to a more robust and resilient IoT ecosystem.

Chapter 2

Literature Review

The literature on IoT security underscores a range of vulnerabilities, particularly in low-power and lossy networks (LLNs)[17] where resource constraints are a critical challenge. A notable security threat in these networks is the rank attack, which exploits routing protocols to manipulate the rank parameter, leading to inefficient routing, increased energy consumption, and even network partitioning.

While various security mechanisms have been proposed to safeguard IoT networks, many focus on broader attacks like Sybil or blackhole attacks[20], with insufficient attention given to the rank attack's specific risks. Traditional detection mechanisms in IoT[1] often rely on complex algorithms that are computationally expensive, making them unsuitable for the energy-constrained nature of IoT devices.

Research shows a growing emphasis on lightweight security solutions[7] that can operate efficiently in such constrained environments. However, current approaches either lack the ability to detect rank manipulation or introduce too much overhead, which can degrade the overall network performance. Moreover, many existing trust-based mechanisms for securing communication in IoT networks[14] focus on general trust establishment between nodes but do not directly address how malicious nodes may exploit the routing process itself.

The TRank-IoT mechanism seeks to address these limitations by introducing an efficient, trust-based solution[7] specifically designed for detecting and mitigating rank attacks in IoT networks. By focusing on a lightweight design, the approach aims to minimize computational and energy overhead, making it practical for deployment in real-world IoT systems.

TRank solution is designed to scale with the network, ensuring that as the number of devices grows, the system remains robust and secure without sacrificing performance. Through this, TRank-IoT aims to fill the critical gap in the literature by providing a specialized, efficient mechanism tailored for detecting rank attacks[5], something that has

been under explored in current research.

2.1 Related Research

2.1.1 A Trust-Based Model for Secure Routing against RPL Attacks in Internet of Things[16]

2.1.1.1 Summary of the research item

Muzammal et al. [16] proposed a secure, Trust-Based model, **SMTrust** designed to mitigate **Rank** and **Blackhole** attacks in RPL-based IoT networks. This model uses various trust metrics including mobility, energy level, success rate, historical analysis, to analyze node behavior and ensure secure routing in fixed and mobile IoT environments.

SMTrust was evaluated using Contiki-2.7/COOJA simulator and topology stability, packet loss rate and the products showed remarkable improvements. The results show that SMTrust outperforms existing trust-based algorithms such as SecTrust and MRTS, with a 46% increase in topology stability, a 45% decrease in packet loss, and a 35% increase in throughput, while only having very little increase in power consumption.

Although **SMTrust** uses a comprehensive approach to trust-based routing but it has some limitations. The authors rely primarily on simulation-based research, which does not consider the full complexity of real-world IoT scenarios. Furthermore, although the model is computationally efficient, it introduces a small overhead due to ongoing reliability analysis, which may affect the energy efficiency of IoT devices.

It is well aligned with our proposed TRank solution, as both aim to reduce Rank attacks on RPL-based networks through a lightweight trust-based mechanism. However, TRank seeks to further extend this work by optimizing detection algorithms specifically in the Contiki-NG operating system and TRank will focus on enhancing scalability and detection accuracy in dynamic environments.

2.1.1.2 Critical analysis of the research item

This Paper [16] provides a comprehensive approach to mitigate routing attacks on RPL networks using a trust-based model called SMTrust. Analysis of the paper reveals several strengths and limitations:

Strengths

1. **Addressing a Critical Problem:** The paper effectively identifies one of the most

critical issues in IoT networks vulnerabilities to various types of attacks such as Rank and Blackhole attacks of the RPL protocol. The authors highlight the high level of reliance on IoT systems and the urgent need for robust security measures, making the study relevant and timely.

2. **Innovative Trust Metrics:** SMTrust includes a variety of trust metrics including historical analysis, success rates, energy levels, and roaming factors, which are important in a dynamic IoT environment this multi-faceted approach enhances the model's ability to analyze node behavior more comprehensively, and improves the reliability of routing decisions.
3. **Performance Evaluation:** The authors provide a comprehensive evaluation of the proposed model through simulations under realistic conditions, showing improved performance parameters such as topology stability, throughput, and reduced packet loss rates while compared to existing protocols for example 46% increase in topology stability and 45% reduction in packet loss, highlights the effectiveness of the SMTrust model.
4. **Robust Methodology:** The paper's methodology is well-structured, with a clear outline of the trust evaluation process, attack detection mechanisms, and performance metrics used for evaluation. The use of Contiki/COOJA for simulation provides a reliable platform for testing, adding credibility to the findings.
5. **Consideration of Node Mobility:** One of the significant contributions of this work is the consideration of node mobility, which is often overlooked in trust-based models. By integrating mobility into the trust evaluation process, the authors enhance the model's applicability to real-world IoT applications where devices are frequently moving.

Limitations

1. **Trust Manipulation through Collusion:** A notable limitation of the SMTrust model is the potential vulnerability to shared attack. If a group of nodes conspires to reduce the trust value of a valid node, SMTrust[16] may incorrectly perceive it as malicious, and automatically isolate itself from the network. The paper acknowledges this distinction however it does not propose solutions or mitigation strategies to deal with allocations, which are critical to managing complex networks.
2. **Complexity and Overhead:** Although the model is designed to be lightweight, the inclusion of multiple trust metrics and the computation of trust indices may introduce computational overhead, especially in resource-constrained environments typical of IoT devices. The paper does not adequately address the potential impact

on energy consumption and processing time, which are critical considerations in the design of IoT systems.

3. **Limited Dynamic Testing:** The simulations primarily focus on static and mobile sender nodes but do not comprehensively assess the model's performance under highly dynamic conditions, such as rapid node movement or frequent topology changes. The lack of evaluation in varying mobility scenarios could limit the model's effectiveness in practical applications where such dynamics are common.
4. **Dependency on Trust Metric Weighting:** The trust evaluation process relies on the correct weighting of trust metrics, which can be subjective. The paper does not provide extensive justification for the chosen weights or explore the implications of varying these weights on the overall trust assessment, leaving a gap in understanding how sensitive the model is to these parameters.

2.1.1.3 Relationship to the proposed research work

The paper[16] "A Trust-Based Model for Secure Routing against RPL Attacks in Internet of Things" is directly related to our proposed research on TRank in several important ways. Firstly, both the paper and TRank focus on increasing the security of routing protocols for low-power lossy (RPL) networks against specific routing attacks, especially Rank and Blackhole attacks.

The findings of this paper as effective on trust based strategies to mitigate such vulnerabilities in the RPL network provides a basic overview of what can be used. Secondly, the paper discusses the importance of incorporating **trust measures** to evaluate node behavior in a network. This idea fits well with TRank's goal of designing a small **trust-based detection mechanism** that evaluates the trustworthiness of nodes to detect and isolate malicious behavior.

The metrics proposed in the paper can be provided the selection of reliability evaluation criteria has been identified and will be used in TRank. Furthermore, the investigation of the evolution of nodes as an important factor in reliability analysis is of particular relevance to the paper. TRank aims to address Rank attacks in dynamic IoT environments, where node mobility is an important concern.

As the authors suggest, incorporating navigation considerations into the **reliability evaluation** process can improve the TRank in real situations. Additionally, the paper highlights limitations, such as the possibility of enforcing trust through shares. This limitation is important to address in TRank, as it can affect the integrity of the reliability assessment process. Understanding these vulnerabilities can guide countermeasures within TRank, ensuring a much stronger defense against collective attacks.

Finally, the simulation results presented in the paper provide a comparative framework for evaluating TRank's performance. By benchmarking against the findings and methodologies outlined in this paper, I can effectively demonstrate the improvements and innovations introduced by TRank in securing RPL networks.

2.1.2 RPL-based routing protocols in IoT applications: A Review [11]

2.1.2.1 Summary of the research item

This paper[11] looks at RPL (IPv6 Routing Protocol for Low-Power and Lossy Networks) and its use in various Internet of Things (IoT) applications. RPL was introduced by the Internet Engineering Task Force (IETF) in 2012 to manage connectivity between low-power devices in resource-limited networks.

The protocol is a **Directed Acyclic Graph** (DAG) hierarchical structure for routing data, using techniques such as Trickle Timer to reduce redundant messages RPL's flexibility allows it to adapt to IoT applications by changing its routing decisions various based on network needs. The paper examines the performance of RPL in key IoT sectors such as healthcare, smart places, transportation, infrastructure, and military applications. In healthcare, RPL supports reliable data connectivity for patient care, but its functionality is reduced in mobile environments such as ambulatory patient tracking.

Smart cities for smart environments RPL scales well for applications such as agriculture, but in transportation systems such as vehicular networks, RPL is not as effective in dealing with mobile devices. RPL's flexibility is useful in industrial and military contexts, however due to the sensitivity of these areas, energy efficiency And security is a major concern.

The study[11] also highlights several challenges with RPL and discusses improvements suggested by researchers. A key issue is **energy efficiency**—while RPL works well on static grids, it is less effective on large dynamic grids. Alternative methods for energy conservation have been proposed, but these tend to complicate the communication process. Mobility is another weakness; RPL struggles with mobile nodes, making it less suitable for dynamic IoT environments such as transportation.

Some solutions, such as mRPL, help manage mobile nodes more efficiently, but can consume a lot of energy. Congestion is another problem in dense networks, where data can be delayed or lost. Strategies have been developed to reduce congestion, but they often add additional costs to the network. Finally, RPL is vulnerable to security threats such as **denial-of-service** attacks because it does not have strong built-in protections. Researchers have proposed several strategies to improve safety, but these should be fully

integrated into RPLs.

RPL is a valuable protocol for IoT. However, challenges are present, especially in terms of traffic, congestion and security management. The paper suggests that future research could focus on improving these areas and that new RPLs may also need to be developed to meet evolving IoT application requirements. Despite the limitations, RPL is an important component in IoT networks, and ongoing research will help to optimize its functionality for different applications.

2.1.2.2 Critical analysis of the research item

The paper provides an insightful review of RPL (IPv6 Routing Protocol for Low-Power and Lossy Networks) and its role in IoT applications, highlighting its strengths and limitations. RPL has proven to be a highly flexible and energy-efficient protocol for low-power IoT environments, making it widely applicable across diverse sectors such as healthcare, smart cities, industry, and military applications. Its hierarchical structure and adaptable objective functions allow it to balance network performance and energy conservation effectively in static networks.

Strengths

1. **Energy Efficiency:** RPL protocols are designed for low-power and lossy networks, making them ideal for IoT devices with limited energy resources. Enhancements like the trickle timer and optimized routing metrics (expected transmission count - ETX) significantly reduce unnecessary control messages, thereby conserving energy.
2. **Scalability and Flexibility:** RPL's design allows it to efficiently manage large networks by building a Directed Acyclic Graph (DAG) and Destination Oriented DAG (DODAG). The flexibility of the Objective Function (OF) makes it adaptable to different network requirements and applications.
3. **Broad Applicability:** RPL has been successfully applied across various domains including healthcare, smart environments, industrial systems, and military applications. This versatility is a major advantage in different IoT scenarios.
4. **Reliability:** RPL ensures reliable data transmission in most static and low-mobility scenarios by using mechanisms like DIO and DAO control messages, which ensure network connectivity and data integrity.

Limitations

1. **Lack of Mobility Support:** One of the major shortcomings highlighted is RPL's struggle with mobile nodes. Although static networks are handled efficiently, the

protocol's performance degrades in highly dynamic environments where nodes frequently change positions. Enhancements such as mRPL and other mobility-aware versions attempt to address this but require further improvements for large-scale practical deployment.

2. **Congestion Issues:** RPL faces congestion problems in dense networks with multiple hops. Congestion can cause delays, dropped packets, and reduced throughput. Several studies have proposed congestion control mechanisms, but there are trade-offs between energy efficiency, scalability, and throughput that still need to be optimized.
3. **High Control Message Overhead:** Despite the trickle algorithm's goal of reducing control messages, in dynamic or high-node-density environments, the control message overhead remains an issue, leading to wasted energy and increased delays.
4. **Security Vulnerabilities:** The paper identifies that the RPL standard does not offer robust in-built security mechanisms. Attacks such as sinkhole, blackhole, and version number attacks are significant threats, and while some improvements have been proposed, there are gaps in securing the protocol against such attacks. Security measures need to be integrated without sacrificing network performance or flexibility.

2.1.2.3 Relationship to the proposed research work

The reviewed paper[11] is highly relevant to my project titled "**A lightweight and trust-based rank attack detection mechanism for IoT networks - TRank**", as it thoroughly explores the Routing Protocol for Low-Power and Lossy Networks (RPL), the core protocol I am working with. RPL is widely used in IoT environments, particularly in wireless sensor networks (WSNs), but the paper points out several security vulnerabilities inherent in RPL, including attacks that exploit the rank-based nature of the protocol. These vulnerabilities are directly linked to the type of attack my project aims to address: **rank attacks**.

In RPL, nodes form a hierarchical structure using ranks, where a node's rank defines its position relative to the root of the network (sink node). A rank attack occurs when malicious nodes misrepresent their rank to either attract more traffic by pretending to be closer to the root or disrupt network communication by misguiding data traffic.

This can lead to severe network inefficiencies, packet loss, increased latency, and compromised energy consumption, which undermines the primary goals of IoT networks. The paper provides a detailed discussion of RPL's security limitations, which include vulnerabilities to attacks like sinkhole attacks, blackhole attacks, and version number attacks, all

of which share common elements with rank attacks. While it explores some mitigation techniques for these attacks, the paper lacks a comprehensive solution for rank attacks.

This gap strengthens the need for the TRank mechanism, which aims to provide a targeted solution to detect and mitigate rank attacks in RPL-based IoT networks.

2.1.3 A Review on the Security of IoT Networks: From Network Layer's Perspective [10]

2.1.3.1 Summary of the research item

This paper provides a comprehensive analysis of security challenges in Internet-of-things (IoT) networks, with a particular focus on the network layer, especially during routing protocols for low-power loss (RPL) networks. As IoT devices become more ubiquitous and increasingly connected, in particular at the network layer, which is susceptible to attacks, the need for robust security measures increases. The study focuses on attacks that are external like sinkhole, selective forwarding, blackhole, sybil, wormhole attacks, which target the network layer and can significantly reduce network performance.

The paper [10] highlights how IoT devices with limited power and constraints are particularly susceptible to these attacks due to limitations in RPL. The authors provide an in-depth discussion of security vulnerabilities in IoT networks, classify attacks and explain their impact on network performance through key parameters such as energy consumption, **packet delivery ratio** (PDR), scalability, throughput, etc.

An important contribution of this paper is the exploration of machine learning based techniques for detecting and mitigating network attacks. The authors provide an overview of various ML algorithms, such as K-Nearest Neighbors (**KNN**), Decision Trees (**DT**), Support Vector Machines (**SVM**), and Random Forest (**RF**), and their applications in IoT network security. Specific ML-based security solutions aimed at detecting threats are also discussed from within such as sinkhole attacks, and by incorporating ML models constrained IoT environments present associated challenges.

In addition to analyzing the attacks, the paper also explores various cryptography tools, **intrusion detection systems** (IDS), and trust-based strategies proposed to secure IoT networks. Sub-cryptography strategies such as AES and present are recommended for IoT with low energy devices, while distributed IDS systems are discussed in terms of the ability to provide dedicated security in IoT networks.

The paper concludes by highlighting several gaps in the current research environment, such as the lack of a standardized, unified approach to use machine learning to detect attacks from humans. This also highlights the need to find scalable, energy-efficient security solutions that can be truly deployed - world IoT systems. Future research directions

suggest exploring more robust ML-based models, distributed security mechanisms, and less destructive cryptographic techniques to enhance security at the network level

This study is a valuable resource for researchers looking to understand the current state of IoT security, especially in the network, and provides insights into the challenges and future research opportunities in securing IoT networks in case of internal and external threats.

2.1.3.2 Critical analysis of the research item

The paper **A Review on the Security of IoT Networks: From Network Layer's Perspective** provides a thorough examination of security challenges in IoT networks, focusing particularly on network layer attacks. Here's a critical analysis based on its contents:

Strengths

1. **Broad Scope:** The paper spans a wide range of issues affecting the network layer, making it a useful resource for researchers and practitioners. It meticulously discusses attacks such as selective forwarding, blackhole, Sybil, and sinkhole, highlighting their impact on network performance.
2. **Relevance of ML Approaches:** The paper does a commendable job of highlighting how machine learning, especially techniques like clustering algorithms, intrusion detection systems (IDS), and anomaly detection models, can address network-layer vulnerabilities. This is a cutting-edge approach, reflecting recent trends in leveraging AI for IoT security.
3. **Evaluation of Algorithms:** Various ML techniques (K-Nearest Neighbors, Support Vector Machines, and Random Forest) are discussed in terms of their effectiveness for detecting specific IoT attacks. The paper critically examines each method's strengths and weaknesses, such as accuracy in detection and the impact on network resources.
4. **Energy Efficiency Discussion:** Energy consumption is a particularly pressing concern in IoT devices due to their limited battery life. The paper rightly emphasizes this issue, noting how attacks like sinkhole or selective forwarding can drain battery resources by causing unnecessary data transmissions. It also mentions how some security measures can add to energy overhead, which is crucial for designing low-power IoT systems.
5. **Exploration of Intrusion Detection Systems (IDS):** The discussion of IDS is valuable, particularly the mention of lightweight IDS that are distributed across nodes, allowing each node to monitor its neighbors. This aligns with the decentralized

nature of IoT systems and shows a good understanding of how to balance security with system limitations.

6. **In-Depth Discussion of Cryptographic Tools:** The paper discusses cryptographic solutions like AES and other algorithms, analyzing their suitability for low-power IoT devices. It emphasizes that lightweight cryptographic techniques are essential for RPL protocol security, which is a key insight for developing resource-efficient security solutions.

Limitations

1. **Limited Depth on Attack Variants:** While the paper lists several types of attacks, its discussion on the variations and subtle differences within each attack type (different mechanisms for selective forwarding or Sybil attacks) is not as deep as expected for a comprehensive review. A more granular breakdown of how each variant can impact IoT networks would enhance the review.
2. **Lack of Practical Feasibility Analysis:** While the paper praises ML approaches for their effectiveness, it underplays the challenges of implementing such techniques on resource-constrained IoT devices. For example, most ML models are computationally expensive, requiring significant processing power and memory. The paper does not adequately address how these limitations would affect the real-world deployment of ML-based solutions in IoT environments where energy and processing power are scarce.
3. **Scalability Concerns:** The discussion of scalability for ML-based solutions is limited. The paper does not provide insights into how well these models perform as the number of IoT devices increases or as networks scale up in size. Real-world IoT systems can include thousands of devices, and a deeper exploration of how ML solutions maintain effectiveness at this scale would have been beneficial.
4. **Vague Recommendations:** While the paper identifies gaps, it lacks specific, actionable research proposals. For example, it mentions the need for distributed ML solutions but doesn't provide clear guidelines on how to develop these models or what specific techniques could be explored further. It would be beneficial to suggest particular ML algorithms, frameworks, or hybrid approaches that could be tested in future studies.
5. **Limited Quantitative Analysis:** Despite mentioning key metrics, the paper lacks a quantitative analysis of how different attacks and countermeasures affect these metrics in real-world scenarios. For instance, it does not provide any benchmarks or empirical data showing the extent to which energy consumption, PDR, or throughput is impacted by specific security attacks.

2.1.3.3 Relationship to the proposed research work

The paper[10] "A Review on the Security of IoT Networks: From Network Layer's Perspective" provides important insights directly related to the solution we have proposed, TRank-IoT. Both the paper and our solutions focus on critical vulnerabilities in IoT system networks, specifically in the Routing Protocol for Low-Power and Lossy Networks (RPL).

Over various RPL protocols, such as sinkhole, internalization of the sybil, **selective forwarding**, and rank attacks, it highlights the serious impact of these attacks on IoT network performance. One of the key issues discussed is rank attack, where malicious nodes change their rank in the RPL topology to mislead routing decisions, and degrade network integrity and security. My proposed solution, TRank-IoT, targets live attacks.

It provides a **trust-based tool** to detect and mitigate rank-manipulation attacks in real-time, while still being lightweight and scalable for IoT environments. Our focus is on rank attack, acknowledged as a major risk in this paper. This design addressing security gaps not adequately covered by current solutions enforces the benefits and importance of my approach especially at the network layer where the RPL protocol operates.

2.1.4 Threats, Detection and Mitigation of Rank Attack: A Survey [13]

2.1.4.1 Summary of the research item

The paper "Threats, Detection and Mitigation of Rank Attack: A Survey" examines security concerns in IoT networks, with special focus on Rank Attacks in RPL protocol, which are essential for effective strategy in Low Power and Lossy networks (LLNs) exploit the rank-based routing of RPLs, which hampers network performance through non-optimized routes, routing loops, and increased control overhead.

The study mainly discusses three types of rank attacks: rank increase, rank decrease, and parent attacks worst case, each affects the quality and security of the network. This is achieved through **exchange-based techniques** (using cryptography, trust models, and boundaries) and various detections classified as intrusion detection systems (IDS).

Research Mitigation Techniques aims to protect networks and restore network performance by identifying malicious nodes, but challenges remain to balance energy consumption, **scalability**, and accuracy, especially in dynamic environments, and the paper concludes by emphasizing the need for comprehensive solutions.

2.1.4.2 Critical analysis of the research item

A detailed critical analysis of paper[13] highlights both its contributions and its limitations in addressing security issues related to rank attacks in IoT networks, particularly those using the RPL protocol.

Strengths

1. **Comprehensive Survey of Rank Attacks:** The paper[13] thoroughly categorizes and explains different types of rank attacks (Increased Rank, Decreased Rank, and Worst Parent), providing readers with a deep understanding of how these attacks exploit the RPL protocol's vulnerabilities. This categorization clarifies the diverse impacts of rank manipulation on network topology, such as un-optimized routes, increased delays, and energy inefficiencies, which are critical in resource-constrained IoT environments.
2. **Extensive Review of Mitigation Techniques:** The survey does an excellent job of reviewing a variety of detection and mitigation strategies. By categorizing them into modification-based and Intrusion Detection Systems (IDS)-based solutions, the paper[13] showcases the breadth of research approaches available. Modification-based techniques include cryptographic methods, trust models, and threshold-based methods that make the RPL protocol more robust. IDS-based techniques rely on identifying anomalies through network behavior monitoring or machine learning algorithms, providing a more dynamic and adaptive defense approach.
3. **Highlighting Open Research Areas:** One of the paper's key contributions is identifying gaps and future research directions. The authors emphasize that most existing solutions are limited to static environments and often focus on one or two attacks at a time. This highlights a critical need for more comprehensive solutions that can handle multiple concurrent attacks in dynamic and mobile environments. Additionally, the paper suggests that machine learning and deep learning approaches could play a significant role in improving the detection accuracy and response time of future solutions.
4. **Detailed Comparison of Defense Mechanisms:** The paper provides a valuable comparative analysis of various mitigation techniques. This includes metrics like energy consumption, detection accuracy, message overhead, and scalability. By doing so, the paper[13] helps researchers and practitioners identify which solutions may be best suited for particular use cases, especially in IoT networks with severe resource constraints.

Limitations

1. **Limited Real-World Evaluation:** While the paper extensively reviews various solutions, many of these approaches are theoretical or have only been tested in simulation environments. This lack of real-world evaluation limits the applicability of these techniques to actual IoT deployments, where network dynamics, interference, and other operational challenges may affect performance. Solutions that work in a controlled lab setting might face difficulties when implemented in real networks with unpredictable conditions.
2. **Focus on Single Attack Types:** Most of the reviewed detection and mitigation methods are designed to handle one type of attack at a time, such as the Increased or Decreased Rank Attack. However, in a real-world scenario, an attacker may launch multiple, simultaneous attacks (e.g., combining rank attacks with wormhole or Sybil attacks), overwhelming the defenses designed for specific threats. The paper acknowledges this limitation but does not delve into any comprehensive approaches that could tackle multi-faceted, complex attacks.
3. **Energy Overhead and Scalability Issues:** Many of the mitigation techniques reviewed in the paper, especially cryptographic methods and trust-based approaches, introduce significant computation and communication overhead. For IoT devices, which are inherently resource-constrained (in terms of power, memory, and processing capacity), these overheads could negatively impact network performance, especially in large-scale deployments. The paper provides a comparison of energy efficiency for some solutions, but more emphasis on scalability in large networks with dynamic nodes would have been beneficial.
4. **Machine Learning Approaches in Early Stages:** Although the paper mentions the potential of machine learning (ML) and deep learning (DL) techniques in detecting rank attacks, the discussion is relatively brief. While there are references to a few ML-based approaches, the field is still in its infancy, and more research is needed to develop robust ML-based detection systems that can adapt to different types of attacks in real-time. Additionally, the high computational requirements of ML techniques pose a challenge for their implementation in resource-constrained IoT devices. The paper could have provided more insights into how these challenges can be mitigated or whether lightweight ML techniques could be developed for IoT.

2.1.4.3 Relationship to the proposed research work

The proposed solution, TRank is working on an important gap found in the paper[13] "**Rank Attack Threat, Detection, and Mitigation: Survey**" like the challenges of mitigation strategies, including high computing costs, inefficient energy efficiency, and limited use of existing solutions in a real-world IoT environment among different types.

Many proposed methods, especially models based on **cryptographic authentication**, although effective, come with significant robustness and performance requirements, making them not suitable for IoT devices with features as it is essential.

TRank addresses this concern with a lightweight design that focuses on reducing power consumption and **computational complexity**, thus ensuring compatibility with compact IoT environments. Incorrect as traditional trust-based methods that can accommodate large amounts of resources, TRank combines trust checking and status verification in a way that Reduces network costs while maintaining security. This is especially important because of the paper's criticism that many existing solutions such as cryptography heavy methods cannot therefore balance security with energy efficiency.

Furthermore, the paper distinguishes between handling multiple types of attacks and protecting dynamic environments, such as mobile nodes, where the topology often changes. TRank mitigates this with a dynamic approach a trust-based continuous analysis of node actions, making it adaptive to changes in the network. Using **real-time reliability analysis**, TRank not only detects rank manipulation but also adapts to changes in network conditions, ensuring robust remediation without sacrificing performance. This positions TRank as a novel contribution that addresses the limitations outlined in the survey, particularly in creating scalable, energy-efficient, and adaptable solutions for rank attack detection in IoT networks.

2.1.5 Impact Analysis of Rank Attack on RPL-Based 6LoWPAN Networks in Internet of Things and Aftermaths [4]

2.1.5.1 Summary of the research item

The paper provides an in-depth analysis of rank attack, which is a serious denial-of-service (DoS), and manipulative threat to RPL-based 6LoWPAN networks commonly used in IoT. RPL is a remote vector routing protocol with low energy. Designed for loss-less networks (LLNs), which form the backbone of IoT communications by **facilitating IPv6 connectivity**, however, the protocol's reliance on rank values assigned to the network nodes to organize the network's routing hierarchy makes it easily adaptable. A rank attack occurs when an attacker node advertises a false rank value, and disrupts the topology. This can lead to inefficient routing, network loops, and severe network performance degradation.

The paper investigates the impact of these attacks on RPL network topologies, including grid random networks with fixed and mobile nodes. The simulation shows that attack packet delivery, inter-packet time, control message overhead, power - and affect key metrics such as its consumption. For example, packet reception at the sink node decreases

by 10-13% for grid topologies and 8% for random topologies under attack. Similarly, the inter-packet time and message overhead increase significantly, leading to higher energy consumption and reduced network performance.

The analysis[4] also revealed that the position of the attacker in the network plays an important role in determining the severity of the attack, with nodes closer to the sink causing more interference. These solutions aim to protect the network from topology manipulation while maintaining the lightweight nature required by resource-constrained IoT devices.

2.1.5.2 Critical analysis of the research item

The paper offers a comprehensive analysis of the rank attack and its detrimental effects on RPL-based 6LoWPAN networks in IoT environments. Below is a critical analysis focusing strengths and limitations:

Strengths

1. **In-Depth Examination of Vulnerabilities:** The paper[4] presents a thorough analysis of the vulnerabilities within the RPL protocol, particularly concerning the manipulation of rank values. It effectively identifies how these vulnerabilities can be exploited through rank attacks, thereby contributing significantly to the understanding of security challenges in RPL-based 6LoWPAN networks.
2. **Diverse Topological Analysis:** The study evaluates the effects of rank attacks across multiple network topologies, including grid and random configurations with static and mobile nodes. This diversity enriches the findings by demonstrating the varying degrees of susceptibility to rank attacks depending on the network's structural characteristics.
3. **Robust Experimental Design:** Utilizing the Cooja simulator, the paper[4] outlines a well-structured experimental setup that replicates real-world IoT scenarios. This methodology enhances the validity of the results, as it closely mirrors the operational conditions of actual 6LoWPAN networks.
4. **Quantitative Performance Metrics:** The paper employs a comprehensive set of performance metrics, including packet delivery ratios, inter-packet times, control message overhead, and power consumption. The quantitative data presented strengthens the argument regarding the detrimental impacts of rank attacks on network performance, providing concrete evidence of the issues at hand.
5. **Insights on Attack Dynamics:** The analysis reveals the critical influence of the attacker node's position within the network topology, offering valuable insights for

developing targeted defensive strategies. This focus on the attacker's strategic positioning lays the groundwork for future research on real-time monitoring and mitigation approaches.

6. **Proposed Mitigation Strategies:** The paper does not only identify vulnerabilities but also proposes several practical countermeasures, such as rank verification and enhancements to objective functions. These suggestions contribute to the broader discourse on securing RPL networks and provide a foundation for future research in this area.

Limitations

1. **Narrow Focus on Single Attacker Models:** The analysis predominantly revolves around scenarios involving a single attacker node. This limitation reduces the applicability of the findings in real-world situations where coordinated attacks with multiple compromised nodes are more likely to occur, potentially leading to a more significant impact on network performance.
2. **Scalability of Proposed Solutions:** While the proposed countermeasures are promising, the paper does not thoroughly evaluate their scalability or effectiveness in larger network deployments. The implications of implementing these solutions in extensive IoT ecosystems, characterized by a high density of devices, remain inadequately addressed.
3. **Simulation-Only Approach:** The reliance on simulated environments for the findings introduces a gap in the empirical validation of the results. Without testing in real-world scenarios, the practical applicability and effectiveness of the proposed solutions could be called into question.
4. **Limited Discussion on Countermeasure Feasibility:** The paper presents potential mitigation strategies but lacks a critical assessment of their feasibility and practicality in real-world applications. Challenges such as the additional computational burden on resource-constrained devices or the potential for increased network traffic due to monitoring processes are not adequately discussed.
5. **Contextual Relevance to Specific IoT Applications:** The analysis does not delve into the implications of rank attacks across different IoT applications and domains. A more nuanced approach that considers the specific requirements and vulnerabilities of various sectors could provide deeper insights into tailored mitigation strategies.
6. **Insufficient Exploration of Dynamic Network Conditions:** The study does not adequately address how dynamic factors, such as node mobility and changing environmental conditions, affect the susceptibility to rank attacks and the effectiveness

of mitigation strategies. Future research could benefit from incorporating these dynamic elements to create a more holistic understanding of network behavior under attack.

2.1.5.3 Relationship to the proposed research work

The findings presented in the paper on the impact of rank attacks on RPL-based 6LoW-PAN networks are highly relevant to our proposed solution, which aims to detect and mitigate such attacks in IoT environments using Contiki-NG OS. By identifying the vulnerabilities within the RPL protocol, particularly regarding rank manipulation, the paper underscores the necessity for robust security mechanisms in IoT networks.

My approach leverages **trust-based mechanisms** to enhance rank verification, addressing the specific weaknesses highlighted in the paper, such as the lack of legitimacy checks for advertised rank values. Additionally, the diverse topological evaluations conducted in the study provide a framework for testing the effectiveness of my solution across different network configurations.

This relationship between the existing research and my proposed methodology emphasizes the importance of developing lightweight and efficient countermeasures that can operate within the **resource-constrained** nature of IoT devices, thereby ensuring the integrity and reliability of communication in RPL-based networks.

2.1.6 A Trust-Based Intrusion Detection System for RPL Networks: Detecting a Combination of Rank and Blackhole Attacks[9]

2.1.6.1 Summary of the research item

The paper "A Trust-Based Intrusion Detection System for RPL Networks: Exploring the Combination of Ranked Black Hole Attacks" introduces the **SRF-IoT framework** in order to improve the security of RPL-based IoT networks against routing attacks RPL, it comes down to the fact that the protocol designed for networks, is vulnerable to rank and blackhole attacks, where malicious nodes trick others into routing traffic through it, dropping packets and causing denial of service (DoS). Traditional security mechanisms struggle to deal with this attack handle, especially things, in compact IoT environments. SRF-IoT combines trust-based routing with an intrusion detection system (IDS) called **SRF-IDS** to detect and mitigate attacks.

The IDS monitors the network, collecting reliability metrics based on packet forwarding behavior. These metrics help nodes to calculate the trust value for their neighbors and choose trusted parents for routing, thus avoiding corruption of nodes. This **authentication**

mechanism is added to the RPL protocol, allowing the network to remain secure without interruption. All the schemes are implemented in Contiki-NG and evaluated using the Whitefield simulation framework.

The system demonstrates excellent performance, achieving a packet delivery ratio (PDR) of 92.8% reducing packet drop by five times, reducing parental change events by three times, all with a minimum cost of a they are created less than 2% of SRF primary power - IoT is a **reliability-based** IDS. Based on its ability to integrate detection, which provides stronger protection than methods that rely on either method alone, the SRF-IDS acts as a separate RPL model, which can monitor nodes close enough not to interfere with normal network traffic.

This makes SRF-IoT a lightweight, scalable and responsive solution to protect IoT networks from routing attacks while maintaining high network performance

2.1.6.2 Critical analysis of the research item

Strengths

1. **Combination of Trust-Based and IDS Approaches:**The SRF-IoT[9] framework combines authentication-based and IDS-based techniques, enabling effective detection and mitigation of rank blackhole attacks compared to methods that rely on only one method This two-way approach this improves the accuracy of malicious node detection
2. **Decentralized Detection:**The trust metric is computed locally by nodes based on information from SRF-IDS, reducing the need for frequent monitoring of entire networks. This decentralization leads to reduced energy consumption and faster decision making, which is important for resource-rich IoT devices.
3. **Low Overhead:**The SRF-IoT scheme results in a minimum packet overhead of less than 2% making it highly efficient for IoT environments. This lightweight configuration ensures that the security appliance does not overburden network resources or compromise performance.
4. **High Packet Delivery Ratio (PDR):**The method showed a significant improvement in PDR (92.8%), indicating the effectiveness of the system in maintaining network performance even during attacks
5. **Reduced Parent Switching:**SRF-IoT reduced parental changes by three, indicating that nodes can establish robust routes, increasing overall network stability and reducing unnecessary energy consumption.

Limitations

1. **Dependency on External IDS:** The system relies on SRF-IDS to calculate the reliability metric, which can result in a single failure. If the IDS detectors break down or fail, the network may be weak, or the reliability meter may be inaccurate.
2. **Trust Metric Granularity:** Based mainly on the packet forwarding behavior, the reliability calculation can be very simple. This can be false positive or negative, especially if attackers mimic normal behavior to avoid detection or if beginning nodes experience temporary packet loss due to network issues
3. **Energy Consumption in Promiscuous Mode:** While this approach reduces the power consumption for conventional nodes, SRF-IDS[9] detectors operate in deviant mode, which can result in high power consumption for their devices, and provide not suitable for long-term use in battery-powered IoT environments
4. **Communication Overhead for IDS Updates:** Although the overall cost is low, the frequent exchange of reliability metrics between the SRF-IDS detector and monitored devices can introduce additional communication costs in dense networks or when frequent updating is required, as it can affect network bandwidth

2.1.6.3 Relationship to the proposed research work

"A Trust-Based Intrusion Detection System for RPL Networks", provides a trust-based system[9] to detect rank attacks in RPL, supporting TRank-IoT's goal of reducing such vulnerabilities directly. Detection and reduction of malicious Nodes themselves are key. This circuit-based approach is in line with TRank-IoT's focus on RPL security in IoT networks with minimal resource consumption, ensuring **efficiency** while providing security is improved

2.1.7 RPL-Based IoT Networks under Decreased Rank Attack: Performance Analysis in Static and Mobile Environments[8]

2.1.7.1 Summary of the research item

The paper "RPL-Based IoT Networks under Decreased Rank Attack: Performance Analysis in Static and Mobile Environments" examines the impact of reduced rank attacks on RPL-based IoT networks in static and mobile environments RPL (IPv6 Routing Protocol for Low-Power and Lossy Networks) protocol is widely used in IoT networks due to its efficiency in dealing with redundant devices, but it is still vulnerable to various types of internal attacks, one of them being rank attack, where an attacker node falsely advertise a lower rank. This tricks other nodes into transmitting data, degrading network performance.

The analysis uses a random directional dynamics model (RDM) in the Cooja simulator that reduces rank on key performance metrics such as packet delivery ratio (PDR), **average end-to-end delay** (AE2ED) and throughput. Tests the impact of attacks Transmission Count (ETX), and Power Consumption (APC). Simulations are performed with nodes (10, 20, 30, and 40), different percentages of attacker nodes (10% 20% 30% 40%), and mobile scenarios, which provide a detailed analysis of how so these attacks affect network productivity.

The results show that reduced rank attacks severely disrupt the routing hierarchy of the network, resulting in lower PDR, end-to-end delay, higher energy consumption, higher throughput, thus these negative effects are more pronounced in mobile environments, where node mobility adds more complexity to routing. The findings highlight the importance of robust security measures to protect IoT networks from highlighting such vulnerabilities, especially as the demand for mobile IoT solutions increases

2.1.7.2 Critical analysis of the research item

Strengths

1. **Comprehensive Analysis of Decreased Rank Attack:**The study provides a detailed performance analysis of attack mitigation in RPL-based IoT networks, examining key metrics[8] such as PDR, AE2ED, throughput, ETX, APC etc. This multi-metric approach makes sense as the whole of the impact of attacks on network performance is provided.
2. **Evaluation in Both Static and Mobile Environments:**Unlike most previous studies that focus only on stationary situations, this paper examines attacks in static and mobile environments. The inclusion of a random directional dynamics model (RDM) highlights the severity of attack effects in mobile systems, making the analysis suitable for dynamic IoT applications
3. **Realistic Mobility Simulation:**The use of Cooja 3.0 and the Random Direction Mobility Model (RDM) improves the accuracy of the simulations. The study acknowledges and addresses the limitations of alternative traffic models such as the random way point model (RWP), adding confidence to the findings.
4. **Diverse Metrics for Thorough Evaluation:**Using a wide range of performance metrics—PDR, AE2ED, throughput, ETX, and APC—provides a comprehensive analysis of how attacks degrade network performance. This provides a comprehensive understanding of the impact of the attack on various network components such as of data transmission, latency, and power consumption.

5. **In-depth Discussion of RPL Vulnerabilities:**The paper highlights RPL's weaknesses in addressing ranking-based attacks, and provides insights that can be used to develop robust IoT security solutions. Studies focusing on unique aspects of the RPL ranking system contribute to efforts which continues to secure IoT networks.

Limitations

1. **No Active Defense Mechanisms Proposed:**Although the paper[8] provides a detailed assessment of the impact of the attacks, there are no proposed defensive mechanisms or mitigation strategies to mitigate attacks in rows. This misses the research findings a limiting their effective use to secure real-world IoT networks.
2. **Static Security Assumptions:**The analysis assumes a fixed number of inbound nodes (10% to 40%) throughout the simulations. It does not take into account variable attack behavior or dynamics in which attackers change strategy during simulation, which can provide a more realistic view of network behavior in real-world threats.
3. **Energy Consumption Focus on Attack:**Although the paper analyzes the use of force (APC) under attack scenarios, it does not analyze the force costs in terms of defensive strategies or the use of possible combat strategies. This raises the question of whether defense new possibilities for unlocking low-power IoT devices.
4. **Simplified Network and Mobility Models:**While the Random Directional Mobility Model (RDM) offers some advantages over the Random Waypoint Model (RWP), it represents a simplified mobility model for Real-world IoT applications, such as vehicular networks or stationary systems drone, may reveal more complex motion patterns that may affect the results.
5. **Limited Discussion of Security Protocol Overheads:**While the study shows the impact of attacks on network performance, it does not address the extraneous effects of possible security measures to mitigate these attacks. Lack of research on how security measures would affect metrics such as the use of PDR, latency, or power leaves a gap in the practical application of the findings.
6. **Attack Detection and Response Not Considered:**The paper does not discuss real-time mitigation[8] strategies for queue attack detection or response. While it results in degradation of network performance, it does not provide any real-time analytics or intrusion detection mechanisms that can mitigate attacks when a network is down

2.1.7.3 Relationship to the proposed research work

Performance Analysis in Static and Mobile Environments highlights the performance impact of rank attacks on RPL networks, providing critical insights for TRank-IoT to map attacks and analyze metrics such as PDR and E2ED for comparison. The paper's analysis in both static and mobile environments helps guide the TRank-IoT framework to ensure robust performance in various IoT applications, especially in dynamic network environments

2.1.8 Rank and Worm hole Attack Detection Model for RPL-Based Internet of Things Using Machine Learning[23]

2.1.8.1 Summary of the research item

"Rank and Wormhole Attack Detection Model for Internet Based on RPL Using Machine Learning". The authors highlight the vulnerabilities of IoT systems, especially RPL-specific attacks such as rank attacks (RA) and **sensor network** (SN) legacy attacks such as wormhole attacks (WHA), which affect privacy and fundamental functioning of these systems, and to overcome this, proposes a small multi-class classification model called MC-MLGBM (Multi-class Machine Learning Gradient Boosting Machine).

Key contributions of the paper include the generation of new datasets with new benign traffic attack scenarios simulated in fixed and mobile IoT environments. The study uses the Cooja simulator to generate network models for attack and benign data for training and testing. The **MC-MLGBM** model performs a multiclass classification on these data to distinguish between normal traffic, RA, and WHA.

The proper selection of parameters improves the performance of the model, and various metrics such as accuracy, precision, recall, and Matthews correlation coefficient (MCC) are used to monitor its performance. The results show a high accuracy (99.7%), precision (99%) and recall (99.7%).

The authors note the limitations of existing detection models that typically do not consider the possibility of movement or focus on a single attack. Addressing RPL-specific SN and property attacks in fixed and mobile settings makes the proposed model stand out as comprehensive and effective.

2.1.8.2 Critical analysis of the research item

Strengths

1. **Lightweight Multiclass Classification:**The Light Gradient Boosting Machine (LGBM)

for multiclass classification is computationally efficient. It uses techniques such as one-stage sampling (GOSS) and exclusive feature bundling (EFB), making it lightweight and suitable for resource-constrained IoT environments with limited computation and memory.

2. **Effective Detection of Multiple Attacks:**The proposed MC-MLGBM model handles both RPL-specific rank attack (RA) and sensor network legacy wormhole attack (WHA). Most existing models focus on detecting a single attack, while this approach detects multiple types of attacks, improving overall network security.
3. **High Performance and Accuracy:**The model achieves a high performance of 99.7% accuracy, 99% accuracy, and 99.7% recall, indicating high performance in accurately classifying normal traffic and predicting attacks in IoT networks.
4. **Customized Dataset:**The authors modeled all benign attacks and developed new data sets specifically for IoT networks based on RPL. This addresses the lack of relevant public datasets in this context, making the solution more targeted and applicable to real-world situations.
5. **Faster Training Time:**The histogram-based LGBM algorithm helps speed up training time compared to other booster algorithms like XGBoost, making it suitable for dynamic IoT environments where timely attack detection is important

Limitations

1. **Dependence on Self-Generated Dataset:**Relying on an automatically generated data set suggests that the generalizability of the model may be limited. Real-world networks may exhibit traffic and attack characteristics that differ from those simulated in a controlled environment.
2. **Potential Overfitting:**It may indicate a possible overfitting of the particular data used to obtain a very high accuracy (99.7%). The model may not work so well with unseen, real-world data without further validation using external datasets.
3. **Computation Costs for Larger Data:**Despite its small size, the computational cost of the model can still be high for large data sets or high-traffic areas. Gradient growth methods, including LGBM, can consume more memory and computational resources as the size of the data set increases.
4. **Absence of Feature Explanation:**While feature selection is an integral part of the process, little attention is paid to teaching or teaching how specific features contribute to model decision making This lack of definition can be a limitation in more flexible IoT systems, where the rationale behind detecting attacks has complexity.

2.1.8.3 Relationship to the proposed research work

paper[23] using machine learning (LightGBM) to detect rank and wormhole attacks in RPL networks is consistent with the TRank-IoT goal of reducing rank attacks in Contiki-NG and better classify node actions and isolate malicious nodes in real time. The paper's[23] focus on highly accurate detection of lightweight objects in resource-constrained environments directly supports TRank-IoT's goal of reducing computational costs and improving RPL security

2.1.9 Detection and Verification of Decreased Rank Attack using Round-Trip Times in RPL-Based 6LoWPAN Networks[21]

2.1.9.1 Summary of the research item

The paper "Reduced Rank Attack Detection and Verification Using Round Trip Time in RPL-Based 6LoWPAN Networks" presents a method to detect and mitigate reduced rank attacks in RPL-based IoT networks using **round-trip time** (RTT) occurs over Low-Power Wireless Personal Area Networks in RPL (Routing Protocol for Low-Power and Lossy Networks), but RPL is vulnerable to internal attacks, particularly rank attack, where Malicious nodes falsely advertise ranks to attract traffic from neighboring nodes, destroying network topology and performance

The proposed method uses RTT[21] to detect and verify the reduced attacks in the queue. The server node, assumed to be superior in terms of energy and computing resources, maintains a whitelist and blacklist of network nodes based on their rank per hop average RTT.

During network operation the server sends ICMP echo requests to all nodes and each RTT average periodically, it's the statistics, that when a node announces a new rank lower than its previous one, the server verifies this information by comparing the current RTT with the expected RTT based on the node's rank If RTT does not match the expected value, the node is flagged as malicious and added to the blacklist. This tool helps isolate and remove compromised nodes from the network, ensuring a secure and reliable path.

The paper[21] investigates the process using the cooja simulator and compares the proposed solution with existing detection methods. The results show improved detection accuracy and lower **false alarm rates** compared to other methods. The study concludes that the proposed method efficiently detects queue reduction attacks with relatively low cost, and.

2.1.9.2 Critical analysis of the research item

Strengths

1. **Efficient Detection Mechanism:** The use of RTT provides an efficient minimal method for reduced attack detection. As RTT is a simple metric that can be calculated with minimal resources, the proposed solution is well suited for resource-constrained IoT environments
2. **Whitelist and Blacklist System:** The server node maintains a whitelist and blacklist of nodes, allowing continuous monitoring of evidence of node behavior. This ongoing evaluation allows the network to dynamically adapt to failed nodes, making it more adaptive and secure over time.
3. **Low False Alarm Rate:** Compared with other methods, the proposed method generates a lower number of false alarms, reducing the chances of maliciously flagging the correct node by mistake and therefore the reliability of the system greater than.
4. **Minimal Communication Overhead:** The method is based on ICMP echo requests and RTT measurements resulting in low communication overhead, which is light in terms of bandwidth and power consumption, this makes this method practical for low power networks.
5. **Compatibility with Existing RPL Networks:** The solution[21] integrates seamlessly into existing RPL-based IoT networks without requiring significant modifications to the underlying protocols. This makes it easy to port to existing systems.

Limitations

1. **Single Point of Failure:** The method relies on a central server node to perform all inspection and validation tasks. If this server node becomes damaged or fails, the entire detection system can become inefficient, risking a single failure.
2. **Dependence on RTT for Detection:** The efficiency of this method depends on the accuracy of the RTT measurement. However, RTT can be affected by network corrosion, interference, or latency unrelated to rank attacks, which can lead to false positives or missed detections in some cases.
3. **Lack of Real-Time Adaptation:** Although the method periodically observes node behavior, it cannot respond to rapidly changing attack conditions in real time. Delays between detection cycles allow attackers to exploit the network before they are detected.

4. **High Energy Requirements for the Server Node:** Although the method of identifying IoT devices is limited, it requires significant computing power resources to maintain the whitelist, blacklist, and RTT statistics of the server node, this can make the solution infeasible for networks with under powered server nodes.

2.1.9.3 Relationship to the proposed research work

the paper's RTT-based trust mechanism[21] provides a straightforward and efficient solution to identify reduced rank attacks that can be incorporated into TRank-IoT. This approach is the framework of TRank-IoT to continuously measure and consume node behavior verified by trust metrics. The lightweight RTT analysis supports the need for TRank-IoT to secure RPL without incurring significant overhead, making it a useful method for managing network performance in IoT environments.

2.2 Synthesis of Literature Work

The current literature on securing IoT environments, especially those using the Routing Protocol for Low-Power and Lossy Networks (RPL) offers value but it also highlights several important gaps. TRank-IoT seeks to fill these gaps by developing a reliable and lightweight mechanism for detecting and mitigating category distortion. This synthesis is based on a review of nine important papers. Trank summarizes the relationship between existing research and innovative approaches. TRank-IoT focus on enhancing RPL security without compromising functionality and Scalability of IoT systems.

Existing research on trust-based RPL security: Literature continues to reveal weaknesses of the RPL network, especially Black hole, Sibyl, Rank and Sinkhole, **Paper 1** focuses on reliability-based model for safe routing with RPL. Attacks in the Internet of Things align directly with TRank-IoT's goals by sensing trust variables. This paper introduces the basic reliability rating system for RPL, improved node reliability analysis However, while this article provides a comprehensive overview, No real-time dynamic trust-based protection mechanism is detected. Solutions that are the focus of TRank-IoT

Similarly, **Paper 2** highlights the inherent weaknesses and security vulnerabilities of RPL. Especially categorical attacks. It summarizes the mechanism of ranking management in IoT networks which TRank-IoT aims to manage on a much broader scale, including real-time detection and rank attack interference. This paper highlights the importance of lightweight detection tool for IoT which TRank-IoT will provide. **Paper 3** also discuss the security challenges that IoT networks face, particularly regarding real-time attacks and highlights the importance of targeting rank attacks more effectively.

Challenges in Energy Efficiency and Scalability: Articles 4 and 5 discuss the key lim-

itations of existing solutions. Especially in terms of energy efficiency and scalability. Although it is safe but it is not suitable for resource-constrained IoT devices. **Paper 4** highlights this challenge by pointing out the high computational costs associated with these methods. This is impractical for real-world IoT deployments. TRank-IoT addresses these concerns by providing a lightweight, energy-efficient design. Combining reliability testing and due diligence in a way that reduces costs while maintaining security. The balance between security and efficiency is a key innovation that helps fill the gaps left by current solutions.

Paper 5 also examines how existing rank attack countermeasures fail to adapt to the dynamic IoT environment. Where the network topology changes frequently due to mobile nodes, TRank is based on these insights. It combines a dynamic approach with analysis of the IoT nodes it creates. It continuously evaluates the reliability of nodes in real time. This allows it to adapt to network conditions without degrading performance. This adaptability ensures that TRank-IoT remains effective in both stationary and mobile IoT environments.

Performance Evaluation and Metrics: Several papers emphasize the need for robust performance evaluation to ensure the effectiveness of proposed solutions. **Paper 6** summarizes a trust-based intrusion detection system for RPL networks, which complements TRank-IoT's goal of real-time detection of malicious nodes. However, a comprehensive analysis of it is lacking. Performance indicators such as packet delivery rate (PDR) and end-to-end delay (E2ED) are critical in evaluating the impact of TRank-IoT address ranking attacks.

Papers 7 further contribute to this evaluation by focusing on rank attacks in static and mobile environments. The use of machine learning techniques as described in **Paper 8** is a promising avenue for improving attack detection, although TRank-IoT does not rely on machine learning. But it borrows the idea of real-time classification and isolating malicious nodes. Ensures timely response to remote management attacks. It improves security while reducing computational costs.

Lightweight, trust-based mechanism: **Paper 9** finally introduces a trust-based RTT (round trip) mechanism for detecting ranked attacks. This aligns closely with TRank-IoT's goal, Providing a simple but powerful way to detect and detect rank manipulation, TRank-IoT incorporates the same trust-based parameters. To ensure minimal impact on IoT network resources, the combination of reliability assessment and straightforward design makes TRank-IoT a practical and scalable solution for network security.

2.3 Research Gap for TRank-IoT

The literature on securing RPL-based IoT networks reveals vulnerabilities to Rank, Black-hole, and Sinkhole attacks, although existing research has identified trust-based mecha-

nisms and cryptographic solutions, but they are often computationally expensive, energy inefficient or lack of real-time dynamic detection capabilities in mobile environments.

Trank-IoT addresses these gaps by offering a lightweight, trust-based mechanism. It is specifically designed to detect and mitigate rank management attacks on RPL networks. Unlike existing methods, TRank-IoT focuses on reducing computational energy costs, while adapting to dynamic IoT environments - TRank-IoT offers performance, scalability and guaranteed real-time protection without compromise. This solution targets the need for efficient, scalable, and adaptable security in resource-constrained IoT devices.

2.4 Analysis Summary of research items

Table 2.1: Literature Review Comparison

Parameters	Authors	Year	Domain	Evaluation Metrics	Benefits	Limitations
A Trust-Based Model for Secure Routing against RPL Attacks in Internet of Things[16]	Syeda Mariam Muzammal et al.	2022	The domain of this paper is IoT security, focusing on trust-based models to secure RPL against routing attacks such as Rank and Black-hole attacks.	Topology Stability, Packet Loss Rate (PLR), Throughput, Power Consumption	Improved Topology Stability, Reduced Packet Loss Rate, Increased Throughput, Minimal Impact on Power Consumption, Incorporation of Node Mobility in Trust Metrics	Vulnerability to Trust Manipulation through Collusion, Computational Overhead Due to Multiple Trust Metrics, Performance in Highly Dynamic Scenarios Not Fully Evaluated
A Review on the Security of IoT Networks: From et al. Network Layer's Perspective[10]	ASMA JAHANGEER et al.	2023	Vulnerabilities in the RPL protocol and solutions, including machine learning, to mitigate network-layer attacks.	Energy Consumption, Scalability, Packet Delivery Ratio (PDR), Throughput, Threshold	Comprehensive Analysis of Network-Layer Security, Focus on Machine Learning Solutions, Clear Identification of Performance Metrics	Limited Depth on Attack Variants, Lack of Novel Contributions, Minimal Experimental Validation, Scalability Concerns for ML Solutions, Absence of Real-World Case Studies
Impact Analysis of Rank Attack on RPL-Based 6LoWPAN Networks in Internet of Things and Aftermaths[4]	Udai Pratap Rao et al	2022	Internet of Things (IoT), specifically focusing on RPL-based 6LoWPAN networks.	Packets Received at Sink Node, Average Inter-packet Time, Control Message Count, Power Consumption	Comprehensive Vulnerability Analysis, Diverse Simulation Scenarios, Quantitative Results, Proposed Mitigation Strategies	Single Attacker Focus, Scalability Concerns, Simulation Environment Limitations, Lack of Contextual Analysis

Table 2.2: Literature Review Comparison

Parameters	Authors	Year	Domain	Evaluation Metrics	Benefits	Limitations
Threats, Detection and Mitigation of Rank Attack: A Survey[13]y	De-Harish Kumar Saini et al	2024	Internet of Things (IoT), Security in RPL	Packet Delivery Ratio (PDR), Control Message Overhead, Energy Consumption, Detection Rate, Latency	Comprehensive coverage of rank attacks, Extensive review of mitigation techniques, Identification of open research areas, Detailed comparison of defense mechanisms.	Limited real-world evaluation, Focus on single attack types, Energy overhead and scalability issues, Lack of consideration for mobility
RPL-based routing protocols in IoT applications: A Review[11]	Harith Kharrufa et al.	2019	RPL-based routing protocols for low-power and lossy networks	Energy Efficiency, Reliability, Flexibility, Robustness, Security	Energy Conservation, Scalability, Flexibility in Applications, Efficient Data Transmission	Vulnerability to Attacks, Limited Mobility Support, Congestion Issues, High Control Message Overhead
A Trust-Based Intrusion Detection System for RPL Networks: Detecting a Combination of Rank and Blackhole Attack[9]s	Philokypros P. Ioulianou et al	2022	IoT Security, RPL Networks	Packet Delivery Ratio (PDR), Packet Drop, Parent Switching	Combines trust-based and IDS-based mechanisms, high PDR (92.8%), low packet overhead (<2%), scalable and efficient for large IoT networks	Relies on SRF-IDS for trust metric calculation, potential energy consumption in promiscuous mode, limited attack coverage

Table 2.3: Literature Review Comparison

Parameters	Authors	Year	Domain	Evaluation Metrics	Benefits	Limitations
RPL-Based Networks under Rank Attack: Performance Analysis in Static and Mobile Environments[8]	Amal Hkiri et al	2024	IoT, RPL, Security	Packet Delivery Ratio (PDR), End-to-End Delay (E2ED), Throughput, Expected Transmission Count (ETX), Average Power Consumption (APC)	Comprehensive analysis of decreased rank attacks in both static and mobile environments, provides insights into node scalability and mobility effects on network performance	No proposed defense mechanisms, simplified mobility model, limited real-time attack detection
Rank and Wormhole Attack Detection Model for RPL-Based Internet of Things Using Machine Learning[23]	F. Zahra et al.	2022	RPL-based routing protocols for low-power and lossy networks	Accuracy, Precision, Recall, Cross-Entropy, Cohen's Kappa, MCC	High detection accuracy, Lightweight model using gradient boosting, Novel dataset creation	Limited dataset for only two types of attacks, Mobility scenarios not fully explored, Potential overfitting due to high accuracy in both training and testing
Detection and Verification of Decreased Rank Attack using Round-Trip Times in RPL-Based 6LoWPAN Networks[21]	Abhay Deep Seth et al	2020	IoT Security, RPL-Based 6LoWPAN Networks	Accuracy, Comparison of RTT, Power Consumption, Packet Delivery Ratio	Efficient detection of decreased rank attacks, Utilizes RTT as a detection method, Low overhead due to reliance on RTT	Performance degrades as the percentage of malicious nodes increases, Assumes the server node cannot be compromised, Focuses only on decreased rank attacks without addressing other types of attacks

Chapter 3

Proposed Approach

The TRank-IoT approach improves the reliability of data exchanges within an IoT network. TRank-IoT operates in several key phases: Initialization, Data Transmission, Data Reception, Trust Calculation, Rank Attack Detection, and Trust Update. Each phase is illustrated in the Figure 3.1.

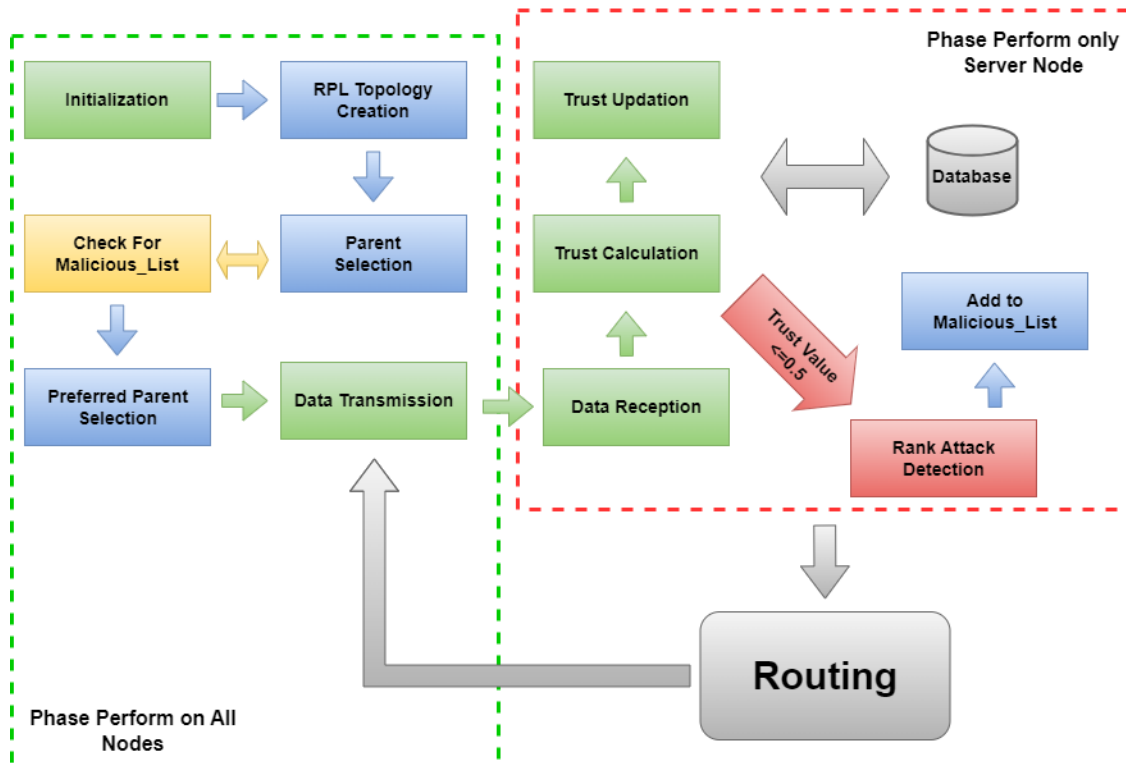


Figure 3.1: System Model of TRank-IoT

3.1 Assumptions:

To establish a clear understanding of the proposed TRank-IoT, the following assumptions were made:

1. Initially, all nodes are authenticated and work cooperatively to construct DODAG tree utilizing the RPL Protocol.
2. All node in network have same energy, processing power, interface, and communication range.
3. All data exchanges are protected by a lightweight, efficient cryptographic method.
4. There is a single server node that is reliable and resource rich.
5. All nodes are either mobile or stationary other than server node.

3.2 Phase Description:

In the **Initialization Phase**, all nodes configure their initial parameters and form a communication structure called DODAG using the Routing Protocol for Low-Power and Lossy Networks (RPL).

Once initialized, the system transitions to the **Data Transmission Phase**. Each node selects a parent node after verifying it against a "Malicious List" to ensure trust. The parent node manages secure data transmissions across the network. This phase focuses on maintaining secure communication by selecting a parent from trusted nodes. Each node collects data from its environment, divides it into packets, and sends them to the Server for processing. These data packets contain key parameters, which the Server uses to calculate the node's trustworthiness.

In the **Data Reception Phase**, all packets received at server node for processing and extracting key information from them.

Simultaneously, during the **Trust Calculation Phase**, the trustworthiness of participating nodes is recalculated based on their behavior and previous communications. Trust values are continuously updated to detect any suspicious behavior, ensuring rapid identification of potential threats.

In the **Rank Attack Detection Phase**, the Server continuously monitors the trust values of nodes present in network, identifying possible rank attacks by analyzing drops in trust levels. If a node's trust value falls below a set threshold, it is flagged as potentially malicious and added to the "Malicious List."

Lastly, in the **Trust Updating Phase**, trust values are revised based on the results of trust recalculations and attack detection. Malicious nodes are added to the "Malicious List," preventing them from participating in future data transmissions. This continuous updating ensures the system's resilience, quickly adapting to changes in node behavior and isolating threats to the network's integrity.

Chapter 4

Conclusions and Future Work

4.1 Conclusion:

The TRank-IoT framework offers a robust and lightweight solution for detecting and mitigating rank attacks within IoT networks. By integrating trust-based mechanisms, TRank-IoT enhances the security and reliability of data exchanges across IoT devices. The proposed methodology efficiently monitors node behavior, dynamically recalculates trust, and isolates potentially malicious nodes in real-time. This adaptive approach ensures the integrity and availability of the network, addressing the critical security challenges posed by rank attacks in resource-constrained IoT environments. TRank-IoT's phases, from initialization to trust updating, work cohesively to safeguard the network from internal threats, ensuring that even in dynamic and potentially hostile environments, the system remains secure and scalable.

4.2 Future Work:

Building on the foundation laid by TRank-IoT, several future directions can be explored to strengthen and validate the proposed solution. Key areas of focus include:

1. **Proposed Solution Enhancement:** Further refinement of the TRank-IoT framework can be explored by optimizing trust calculation mechanisms and enhancing detection accuracy. Improvements in algorithm efficiency will be critical for better performance in large-scale IoT environments.
2. **Implementation of Attack Simulation:** To comprehensively evaluate TRank-IoT's effectiveness, future work will involve implementing and simulating rank attacks in Contiki-NG/Cooja simulator. This will help verify how effectively the proposed solution detects and mitigates rank-based threats.

3. **Implementation of the Proposed Solution:** The next phase involves implementing TRank-IoT on Contiki-NG OS. This will allow us to analyze how well the proposed solution performs particularly with respect to resource constraints and communication overhead.
4. **Simulation and Testing:** Simulating the network environment under various conditions such as varying node mobility, network density, and trust thresholds will provide insights into the scalability and resilience of TRank-IoT. Testing will be conducted using Contiki-NG/ Cooja Simulator to evaluate its performance under different attack scenarios and network configurations.
5. **Comparison with Base Paper:** A critical part of future work will be to compare the results of TRank-IoT with the baseline approaches discussed in the relevant literature, including the solutions proposed in the base papers. Key metrics such as detection accuracy, network overhead, energy consumption, and overall security will be used for comparison.
6. **Final Report:** Following the implementation and evaluation phases, a comprehensive final report will be prepared. This report will include detailed findings, performance analyses, and a discussion on the strengths and limitations of the TRank-IoT framework.

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